Selection Games and Minimal Usco Maps

C. Caruvana

Indiana University Kokomo

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Notation

We will use

- $\mathrm{Fn}(X, Y)$ to denote the collection of partial functions $X \to Y$. For $f \in \mathrm{Fn}(X, Y)$, $\mathrm{dom}(f)$ is the domain of f.
- $\mathrm{gr}(f)$ to denote the graph of f.
- $\mathrm{cl}\,\mathrm{gr}(f)$ to denote the closure of the graph of $f : X \to Y$ in the product topology on $X \times Y$.
- $\mathbf{1}_A$ to denote the indicator function on A; that is, the function that takes value 1 for points in A and 0 otherwise.
- $\mathcal{O}_X$ to denote the collection of non-trivial open covers of X; that is, open covers $\mathscr{U}$ of X with $X \notin \mathscr{U}$.

We will also work with ideals of closed sets, also known as bornologies. Of primary focus are

- $[X]^{<\omega}$, the collection of non-empty finite subsets of X.
- $K(X)$, the collection of non-empty compact subsets of X.

We will also let $\mathbb{K}(X)$ denote the set $K(X)$ endowed with the Vietoris topology.

Characterizing Minimal Usco Maps

- A set-valued function $\Phi : X \to \wp(Y)$ is *upper semicontinuous* if

$$\Phi^{\leftarrow}(V) := \{x \in X : \Phi(x) \subseteq V\}$$

  is open for every open $V \subseteq Y$.
- A set-valued function $\Phi : X \to \wp(Y)$ is *usco* (upper semicontinuous, compact-valued) if $\Phi(x) \in K(Y)$ for all $x \in X$.
- An usco map is *minimal* if its graph is minimal with respect to $\subseteq$.

Let $MU(X)$ be the collection of minimal usco maps from a space X to the space of compact subsets of $\mathbb{R}$. We will consider two primary topologies on this set:

- $MU_p(X)$ is $MU(X)$ with the topology of pointwise convergence and
- $MU_k(X)$ is the $MU(X)$ with the topology of uniform convergence on compact subsets of X.

In general, $MU_{\mathcal{A}}(X)$ is $MU(X)$ with the topology of uniform convergence on members of $\mathcal{A}$ where $\mathcal{A}$ is an ideal of closed subsets of X.

- For a set-valued function $\Phi : X \to \wp(Y)$, a *selection* is a function $f : X \to Y$ so that $f(x) \in \Phi(x)$ for every $x \in X$.
- A function $f \in \mathrm{Fn}(X, Y)$ is *densely defined* if $\mathrm{dom}(f)$ is dense in X.
- A set A is *semi-open* if $A \subseteq \mathrm{cl}\,\mathrm{int}(A)$ and a function $f : X \to Y$ is *quasicontinuous* if $f^{-1}(V)$ is semi-open for all open $V \subseteq Y$. If f is densely defined, we will say it is quasicontinuous if it's quasicontinuous on its domain with the relative topology.
- A function $f : X \to Y$ is *subcontinuous* if, for any net $x_\lambda \to x$, $\langle f(x_\lambda) \rangle$ has an accumulation point.

For $f \in \mathrm{Fn}(X, Y)$, define

$$\overline{f}(x) = \{y \in Y : \langle x, y \rangle \in \mathrm{cl}\,\mathrm{gr}(f)\}.$$

Theorem (Holá & Holý 2009, 2014)

Suppose Y is regular and $\Phi : X \to \wp(Y) \setminus \{\emptyset\}$. TFAE:

- Φ is minimal usco.
- Every selection f of Φ is subcontinuous, quasicontinuous, and $\Phi = \overline{f}$.
- There exists a selection f of Φ which is subcontinuous, quasicontinuous, and $\Phi = \overline{f}$.
- There exists a densely defined selection f of Φ which is subcontinuous, quasicontinuous, and $\Phi = \overline{f}$.

Selection Principles

Given an ideal $\mathcal{A}$ of closed subsets of a space X, $\mathcal{O}_X(\mathcal{A})$ is the collection of $\mathcal{A}$-covers of X; that is, all $\mathscr{U} \in \mathcal{O}_X$ with and,

$$(\forall A \in \mathcal{A})(\exists U \in \mathscr{U})\ A \subseteq U.$$

Note that

- $\mathcal{O}_X([X]^{<\omega})$ is the collection of ω-covers of X, which we will also denote by Ω_X.
- $\mathcal{O}_X(K(X))$ is the collection of k-covers of X, which we will also denote by $\mathcal{K}_X$.

We also define *large covers*: $\Lambda_X(\mathcal{A})$ is the set of all $\mathscr{U} \in \mathcal{O}_X(\mathcal{A})$ so that $\{U \in \mathscr{U} : A \subseteq U\}$ is infinite for every $A \in \mathcal{A}$.

For a space X and an ideal $\mathcal{A}$ of closed sets of X,

- $\mathscr{T}_X$ is all proper non-empty open subsets of X.
- $\mathscr{N}_{X,x} = \{U \in \mathscr{T}_X : x \in U\}$ for $x \in X$.
- $\mathscr{N}_X(A) = \{U \in \mathscr{T}_X : A \subseteq U\}$ for $A \subseteq X$.
- $\mathscr{N}_X[\mathcal{A}] = \{\mathscr{N}_X(A) : A \in \mathcal{A}\}$.
- CD_X is the set of all closed discrete subsets of X.
- $\mathscr{D}_X$ is the set of all dense subsets of X.
- $\Omega_{X,x} = \{A \subseteq X : x \in \mathrm{cl}(A)\}$ for $x \in X$.

The Single-Selection Principle and Game

Definition

$\mathsf{S}_1(\mathcal{A}, \mathcal{B}) := \forall A \in \mathcal{A}^\omega \ \exists x \in \prod_{n\in\omega} A_n \ (\{x_n : n \in \omega\} \in \mathcal{B})$

Definition

$\mathsf{G}_1(\mathcal{A}, \mathcal{B})$ is a game of two players, One and Two.

- In the nth inning, One plays $A_n \in \mathcal{A}$ and Two responds with $x_n \in A_n$.
- If $\{x_n : n \in \omega\} \in \mathcal{B}$, then Two wins. Otherwise, One wins.

Let $\mathbb{P}_X = \{\mathscr{N}_{X,x} : x \in X\}$; for any collection $\mathcal{S}$, let $\neg\mathcal{S}$ be the complement of $\mathcal{S}$. The style of games we'll work with:

- $\mathsf{G}_1(\mathcal{O}_X, \mathcal{O}_X)$, the Rothberger game.
- $\mathsf{G}_1(\mathbb{P}_X, \neg\mathcal{O}_X)$, a game which is isomorphic to the point-open game.
- $\mathsf{G}_1(\mathscr{N}_{X,x}, \neg\Omega_{X,x})$, the clustering variant of Gruenhage's W-game at $x \in X$.
- $\mathsf{G}_1(\mathscr{T}_X, \neg\Omega_{X,x})$ at $x \in X$ and $\mathsf{G}_1(\mathscr{T}_X, \mathrm{CD}_X)$, games of Tkachuk.
- $\mathsf{G}_1(\mathscr{D}_X, \Omega_{X,x})$, the strong countable dense fan-tightness game at $x \in X$.
- $\mathsf{G}_1(\Omega_{X,x}, \Omega_{X,x})$, the strong countable fan-tightness game at $x \in X$.

Full-information Strategies

- A *winning strategy for One* in $\mathsf{G}_1(\mathcal{A}, \mathcal{B})$ is a function $\sigma : (\bigcup \mathcal{A})^{<\omega} \to \mathcal{A}$ so that, whenever $x_n \in \sigma\langle x_m : m < n\rangle$ for all $n \in \omega$, $\{x_n : n \in \omega\} \notin \mathcal{B}$.
  - If One has a winning strategy, we write $\mathrm{I} \uparrow \mathsf{G}_1(\mathcal{A}, \mathcal{B})$.
- A *winning strategy for Two* in $\mathsf{G}_1(\mathcal{A}, \mathcal{B})$ is a function $\tau : \mathcal{A}^{<\omega} \to \bigcup \mathcal{A}$ so that, whenever $A_n \in \mathcal{A}$ for all $n \in \omega$, $\{\tau\langle A_m : m < n\rangle : n \in \omega\} \in \mathcal{B}$.
  - If Two has a winning strategy, we write $\mathrm{II} \uparrow \mathsf{G}_1(\mathcal{A}, \mathcal{B})$.

Limited-information Strategies

- A *predetermined strategy* for One is a strategy which only considers the current turn number. Formally it is a function $\sigma : \omega \to \mathcal{A}$.
  - If One has a winning predetermined strategy, we write $\mathrm{I} \underset{\text{pre}}{\uparrow} \mathsf{G}_1(\mathcal{A}, \mathcal{B})$.
- A *Markov strategy* for Two is a strategy which only considers the most recent move of player One and the current turn number. Formally it is a function $\tau : \mathcal{A} \times \omega \to \bigcup \mathcal{A}$.
  - If Two has a winning Markov strategy, we write $\mathrm{II} \underset{\text{mark}}{\uparrow} \mathsf{G}_1(\mathcal{A}, \mathcal{B})$.

Transfer of Strategic Information

Definition

For two selection games $\mathcal{G}$ and $\mathcal{H}$, we define $\mathcal{G} \leqslant_{\mathrm{II}} \mathcal{H}$ to be the conjunction

- $\mathrm{II} \underset{\text{mark}}{\uparrow} \mathcal{G} \implies \mathrm{II} \underset{\text{mark}}{\uparrow} \mathcal{H}$
- $\mathrm{II} \uparrow \mathcal{G} \implies \mathrm{II} \uparrow \mathcal{H}$
- $\mathrm{I} \not\uparrow \mathcal{G} \implies \mathrm{I} \not\uparrow \mathcal{H}$
- $\mathrm{I} \underset{\text{pre}}{\not\uparrow} \mathcal{G} \implies \mathrm{I} \underset{\text{pre}}{\not\uparrow} \mathcal{H}$

This $\leqslant_{\mathrm{II}}$ is transitive and two selection games $\mathcal{G}$ and $\mathcal{H}$ are *equivalent*, denoted

$$\mathcal{G} \equiv \mathcal{H},$$

if and only if $\mathcal{G} \leqslant_{\mathrm{II}} \mathcal{H}$ and $\mathcal{H} \leqslant_{\mathrm{II}} \mathcal{G}$.

Observe that

$$\begin{aligned} \mathrm{II} \underset{\text{mark}}{\uparrow} \mathsf{G}_1(\mathcal{A},\mathcal{B}) &\implies \mathrm{II} \uparrow \mathsf{G}_1(\mathcal{A},\mathcal{B}) \\ &\implies \mathrm{I} \not\uparrow \mathsf{G}_1(\mathcal{A},\mathcal{B}) \\ &\implies \mathrm{I} \underset{\text{pre}}{\not\uparrow} \mathsf{G}_1(\mathcal{A},\mathcal{B}) \end{aligned}$$

and that $\mathrm{I} \underset{\text{pre}}{\not\uparrow} \mathsf{G}_1(\mathcal{A},\mathcal{B})$ implies a Lindelöf-like principle: for any $A \in \mathcal{A}$, there is $B \in [\mathcal{A}]^{\leqslant\omega}$ so that $B \in \mathcal{B}$.

Theorem (C. & Holshouser, 2019)

Let $\mathcal{A}$, $\mathcal{B}$, $\mathcal{C}$, and $\mathcal{D}$ be collections. Suppose there are functions

- $\overleftarrow{T}_{\mathrm{I},n} : \mathcal{B} \to \mathcal{A}$ and
- $\overrightarrow{T}_{\mathrm{II},n} : \bigcup \mathcal{A} \times \mathcal{B} \to \bigcup \mathcal{B}$

for each $n \in \omega$ so that

1. $x \in \overleftarrow{T}_{\mathrm{I},n}(B) \implies \overrightarrow{T}_{\mathrm{II},n}(x, B) \in B$ and
2. $x \in \prod_{n\in\omega} \overleftarrow{T}_{\mathrm{I},n}(B_n) \wedge \{x_n : n \in \omega\} \in \mathcal{C}$
$$\implies \left\{\overrightarrow{T}_{\mathrm{II},n}(x_n, B_n) : n \in \omega\right\} \in \mathcal{D}.$$

Then $\mathsf{G}_1(\mathcal{A}, \mathcal{C}) \leqslant_{\mathrm{II}} \mathsf{G}_1(\mathcal{B}, \mathcal{D})$.

Game Equivalences and Consequences

X is $\mathcal{A}$-*normal* if, for every $A \in \mathcal{A}$ and open U with $A \subseteq U$, there is an open set V with $A \subseteq V \subseteq \mathrm{cl}(V) \subseteq U$.

Theorem (2022)

Let X be regular and $\mathcal{A}$ and $\mathcal{B}$ be ideals of closed subsets of X. Then

- $\mathsf{G}_1(\mathcal{O}_X(\mathcal{A}), \Lambda_X(\mathcal{B})) \leqslant_{\mathrm{II}} \mathsf{G}_1(\Omega_{MU_{\mathcal{A}}(X),\mathbf{0}}, \Omega_{MU_{\mathcal{B}}(X),\mathbf{0}})$.
- $\mathsf{G}_1(\Omega_{MU_{\mathcal{A}}(X),\mathbf{0}}, \Omega_{MU_{\mathcal{B}}(X),\mathbf{0}}) \leqslant_{\mathrm{II}} \mathsf{G}_1(\mathscr{D}_{MU_{\mathcal{A}}(X)}, \Omega_{MU_{\mathcal{B}}(X),\mathbf{0}})$.
- if X is $\mathcal{A}$-normal, $\mathsf{G}_1(\mathscr{D}_{MU_{\mathcal{A}}(X)}, \Omega_{MU_{\mathcal{B}}(X),\mathbf{0}}) \leqslant_{\mathrm{II}} \mathsf{G}_1(\mathcal{O}_X(\mathcal{A}), \Lambda_X(\mathcal{B}))$.

In this talk, game duality incorporates all the levels of strategy introduced.
By Clontz (2020),

Lemma

Let $\mathcal{A}$ be an ideal of closed subsets of a space X and $\mathcal{B}$ be a collection.

- $\mathsf{G}_1(\mathcal{O}_X(\mathcal{A}), \mathcal{B})$ is dual to $\mathsf{G}_1(\mathscr{N}_X[\mathcal{A}], \neg\mathcal{B})$.
- $\mathsf{G}_1(\mathscr{D}_X, \mathcal{B})$ is dual to $\mathsf{G}_1(\mathscr{T}_X, \neg\mathcal{B})$.
- $\mathsf{G}_1(\Omega_{X,x}, \mathcal{B})$ is dual to $\mathsf{G}_1(\mathscr{N}_{X,x}, \neg\mathcal{B})$.

X is *functionally $\mathcal{A}$-normal* if, for every $A \in \mathcal{A}$ and open U with $A \subseteq U$, there is a continuous $f : X \to \mathbb{R}$ so that $f[A] = \{0\}$ and $f[X \setminus U] = \{1\}$.

Corollary

If X is functionally $\mathcal{A}$-normal, then

$$\begin{aligned}\mathcal{G} := \mathsf{G}_1(\mathcal{O}_X(\mathcal{A}), \mathcal{O}_X(\mathcal{B})) &\equiv \mathsf{G}_1(\Omega_{MU_{\mathcal{A}}(X),\mathbf{0}}, \Omega_{MU_{\mathcal{B}}(X),\mathbf{0}}) \\ &\equiv \mathsf{G}_1(\Omega_{C_{\mathcal{A}}(X),\mathbf{0}}, \Omega_{C_{\mathcal{B}}(X),\mathbf{0}}) \\ &\equiv \mathsf{G}_1(\mathscr{D}_{MU_{\mathcal{A}}(X)}, \Omega_{MU_{\mathcal{B}}(X),\mathbf{0}}) \\ &\equiv \mathsf{G}_1(\mathscr{D}_{C_{\mathcal{A}}(X)}, \Omega_{C_{\mathcal{B}}(X),\mathbf{0}}),\end{aligned}$$

Corollary (continued...)

$$\begin{aligned}\mathcal{H} := \mathsf{G}_1(\mathscr{N}_X[\mathcal{A}], \neg\mathcal{O}_X(\mathcal{B})) &\equiv \mathsf{G}_1(\mathscr{N}_{MU_{\mathcal{A}}(X),\mathbf{0}}, \neg\Omega_{MU_{\mathcal{B}}(X),\mathbf{0}}) \\ &\equiv \mathsf{G}_1(\mathscr{N}_{C_{\mathcal{A}}(X),\mathbf{0}}, \neg\Omega_{C_{\mathcal{B}}(X),\mathbf{0}}) \\ &\equiv \mathsf{G}_1(\mathscr{T}_{MU_{\mathcal{A}}(X)}, \neg\Omega_{MU_{\mathcal{B}}(X),\mathbf{0}}) \\ &\equiv \mathsf{G}_1(\mathscr{T}_{C_{\mathcal{A}}(X)}, \neg\Omega_{C_{\mathcal{B}}(X),\mathbf{0}}),\end{aligned}$$

and the games $\mathcal{G}$ and $\mathcal{H}$ are dual.

If $A \subseteq X$ is sequentially compact and $\Phi \in MU(X)$, then $\Phi[A]$ is bounded.

Theorem (2022)

If $\mathcal{A}$ is an ideal of closed subsets of X, $\mathcal{B}$ is an ideal of closed subsets of X consisting of sequentially compact sets, and X is $\mathcal{A}$-normal,

$$\mathsf{G}_1(\mathscr{N}_X[\mathcal{A}], \neg\Lambda_X(\mathcal{B})) \leqslant_{\mathrm{II}} \mathsf{G}_1(\mathscr{T}_{MU_{\mathcal{A}}(X)}, \mathrm{CD}_{MU_{\mathcal{B}}(X)}).$$

Consequently,

$$\begin{aligned}\mathsf{G}_1(\mathscr{N}_X[\mathcal{A}], \neg\mathcal{O}_X(\mathcal{B})) &\equiv \mathsf{G}_1(\mathscr{N}_{MU_{\mathcal{A}}(X),\mathbf{0}}, \neg\Omega_{MU_{\mathcal{B}}(X),\mathbf{0}}) \\ &\equiv \mathsf{G}_1(\mathscr{T}_{MU_{\mathcal{A}}(X)}, \neg\Omega_{MU_{\mathcal{B}}(X),\mathbf{0}}) \\ &\equiv \mathsf{G}_1(\mathscr{T}_{MU_{\mathcal{A}}(X)}, \mathrm{CD}_{MU_{\mathcal{B}}(X)}).\end{aligned}$$

For a partially ordered set $(\mathbb{P}, \leqslant)$ and collections $\mathcal{A}, \mathcal{B} \subseteq \mathbb{P}$ so that, for every $B \in \mathcal{B}$, there exists some $A \in \mathcal{A}$ with $B \subseteq A$, we define the *cofinality of $\mathcal{A}$ relative to $\mathcal{B}$* by

$$\begin{aligned}&\operatorname{cof}(\mathcal{A}; \mathcal{B}, \leqslant)\\ &\quad = \min\{\kappa \in \mathrm{CARD} : (\exists \mathscr{F} \in [\mathcal{A}]^{\kappa})(\forall B \in \mathcal{B})(\exists A \in \mathscr{F})\ B \subseteq A\}\end{aligned}$$

where CARD is the class of cardinals.

Then, for an ideal $\mathcal{A}$ of closed subsets of a T_1 space X,

$$\operatorname{cof}(\mathcal{A}; \mathcal{A}, \subseteq) \leqslant \omega$$

if and only if $MU_{\mathcal{A}}(X)$ is metrizable.

Corollary

For a regular space X, TFAE:

- X is hemicompact.
- $MU_k(X)$ is metrizable.
- $MU_k(X)$ is not discretely selective.
- $\mathrm{II} \underset{\text{mark}}{\uparrow} \mathsf{G}_1(\mathcal{K}_X, \mathcal{K}_X)$.
- $\mathrm{II} \underset{\text{mark}}{\uparrow} \mathsf{G}_1(\Omega_{MU_k(X),\mathbf{0}}, \Omega_{MU_k(X),\mathbf{0}})$.
- $\mathrm{II} \underset{\text{mark}}{\uparrow} \mathsf{G}_1(\mathscr{D}_{MU_k(X)}, \Omega_{MU_k(X),\mathbf{0}})$.

Corollary

For a regular space X, TFAE:

- X is k-Rothberger.
- $\mathrm{I} \not\uparrow \mathsf{G}_1(\mathcal{K}_X, \mathcal{K}_X)$.
- $MU_k(X)$ has strong countable fan-tightness at $\mathbf{0}$.
- $MU_k(X)$ has strong countable dense fan-tightness at $\mathbf{0}$.
- $\mathrm{II} \underset{\text{mark}}{\not\uparrow} \mathsf{G}_1(\mathscr{N}_X[K(X)], \neg\mathcal{K}_X)$.
- $\mathrm{II} \not\uparrow \mathsf{G}_1(\mathscr{N}_X[K(X)], \neg\mathcal{K}_X)$.
- $\mathrm{II} \underset{\text{mark}}{\not\uparrow} \mathsf{G}_1(\mathscr{T}_{MU_k(X)}, \mathrm{CD}_{MU_k(X)})$.
- $\mathrm{II} \not\uparrow \mathsf{G}_1(\mathscr{T}_{MU_k(X)}, \mathrm{CD}_{MU_k(X)})$.
- $\mathrm{II} \underset{\text{mark}}{\not\uparrow} \mathsf{G}_1(\mathscr{N}_{MU_k(X),\mathbf{0}}, \neg\Omega_{MU_k(X),\mathbf{0}})$.
- $\mathrm{II} \not\uparrow \mathsf{G}_1(\mathscr{N}_{MU_k(X),\mathbf{0}}, \neg\Omega_{MU_k(X),\mathbf{0}})$.

Thank You!